

Howard Wei

howardwei21@gmail.com | hwei.vercel.app | linkedin.com/in/weihoward | github.com/HowardWei123

EDUCATION

San Jose State University

San Jose, CA

B.S. in Computer Science, GPA: 3.69

Expected Graduation: May 2027

Relevant Coursework: Data Structures and Algorithms, Information Security, Operating Systems

EXPERIENCE

Cytokinetics

South San Francisco, CA

Software Engineer Intern

May 2025 – Present

- Engineered a **HIPAA-compliant CrewAI** multi-agent architecture via **LangChain** to perform risk analysis on clinical data systems.
- Developed an end-to-end agentic AI data dashboard application with **AWS** (Bedrock, Lambda, S3) and **Python**, used by hundreds of company scientists to create multiple visualizations at once for millions of data points.
- Created a **Spotfire** IronPython-JavaScript API system to generate advanced scientific visualizations for complex chemical structures, reducing application load times by **50%**.
- Constructed an asynchronous **MQTT** (Message Queuing Telemetry Transport) messaging layer to handle long-running data retrieval and visualization workflows while streaming results to a **Vue.js** frontend.
- Integrated a **PostgreSQL**-backed memory layer to preserve user-agent context across concurrent sessions.

Google Developer Group

Merced, CA

Software Engineering Lead Developer

Sept 2024 – Dec 2024

- Deployed a paper trading platform for analyzing trading habits and providing retail investors with insights through **AWS** (EC2, RDS, Lambda).
- Recruited and directed a **25-member** ML team using **Agile** methodologies (Kanban) and **Jira**, owning **CI/CD** pipelines that automated model testing and deployment.
- Implemented a data **ETL** pipeline to Llama 3 70B API through **Django REST** and **MySQL** market data sources, improving financial advice accuracy by **35%** across 7 different metrics.

Machine Learning at SJSU

San Jose, CA

Software Engineer

Sept 2024 – May 2025

- Developed **TypeScript** API endpoints for club website to store and retrieve user data from **PostgreSQL**.
- Built frontend with **React** and **Tailwind CSS**, allowing member content sharing and event publishing.

PROJECTS

SpotlightAI – Python, GCP, ChromaDB, FastAPI, Postman, GeoPy, PostgreSQL, Streamlit

- Organized development of an **end-to-end system architecture** of a review-based restaurant recommendation RAG model with additional location-based capabilities.
- Integrated **ChromaDB** with recent Google Places API data and historical Yelp reviews to enable retrieval-augmented generation for Gemini, improving response accuracy and reducing hallucinations.

EmotionAI – Python, PyTorch, OpenCV, Pandas, Matplotlib, Streamlit

- Led 5 student engineers across 4 universities in developing a Python-OpenCV human emotion detection model.
- Deployed on **Streamlit** with user-AI interactions through webcam, achieving **60%+** accuracy in live testing.

ByTech – Next.js, Tailwind CSS, TypeScript, Firebase, Pyodide, Agile, Vercel

- Constructed a full-stack online classroom platform for programming teachers, implementing an interactive **Next.js** user interface with **Firebase** data management.
- Leveraged **Agile** methodologies (Scrums), ensuring iterative improvements and regular feedback integration.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, C++, MATLAB, R, SQL, Bash

Frameworks: LangChain, CrewAI, PyTorch, Streamlit, Django, React, Vue, Tailwind CSS

Developer Tools: AWS, GCP, Azure, Git, Jira, Docker, Firebase, Postman

Libraries: ChromaDB, OpenCV, Keras, Matplotlib, Seaborn, NumPy, Pandas, Scikit, Pyodide, SWR

Interests: Bioinformatics, Data Analytics, Formula 1, LLM Inference, Rock Climbing, Systems Architecture